Assignment No 2

Assign 1 : Basic Linux Commands		
Please find the commands for the following. (Find appropriate command line options for the commands and write commands solutions into your notebook)		
No.	**Commands**	**To Do**
1.	man	Command to get help about commands
2.	uname	Print System information
3.	pwd	Print current directory
4.	hostname	Print IP address and hostname of the machine
5.	date	Find today's date
6.	cal	Print calendar for current year
7.	ls, dir	See directory structure
8.	cat	Create file
9.	cat >>	Append some text into existing file
10.	cp	Copy a file, Copy a directory
11.	mv	Move the files
12.	mv	Rename a file
13.	tac, rev	Print the file in reverse order, Reverse each line in the file
14.	wc	Count the words, characters and lines in a file
15.	mkdir	Create directory
16.	cd	Change current directory
17.	rm	Remove a file, remove a directory (with -R)
18.	rmdir	remove empty directory
19.	touch	change the modification time , Create Multiple empty files at once
20.	sort	Sort file.
21.	uniq	Avoid repetition in a file.
22.	head	Print the specified number of lines from top in the file.
23.	tail	Print the specified number of lines from bottom in the file.
24.	factor	Find the factors of a number
25.	alias	Create command shortcut
26.	unalias	remove the command shortcut
27.	echo	Print environment variables individually
28.	ps	Print current processes
29.	kill	Kill the process
30.	env	Print environment variables

Assign 1 : Basic Linux Commands

1. man ==> Command to get help about commands

```
man ls
man cp
man mkdir
man grep
man passwd
```

2. uname ==> Print System information

```
uname
uname -a
uname -r
uname -m
uname -s
```

3. pwd ==> Print current directory

```
pwd
pwd -L
pwd -P
echo $(pwd)
pwd && ls
```

4. hostname ==> Print IP address and hostname of the machine

```
hostname
hostname -I
hostname -f
hostname -A
hostnamectl
```

5. date ==> Find today's date

```
date
date "+%d-%m-%Y"
date "+%A"
date "+%T"
date -u
```

6. cal ==> Print calendar for current year

```
cal
cal 2026
cal 6 2026
cal -3
cal -y
```

7. ls, dir

```
ls
ls -l
ls -a
ls -lh
dir
```

8. cat ==> Create file

```
cat > file1.txt
cat file1.txt
cat file1.txt file2.txt
cat -n file1.txt
cat << EOF > test.txt
```

9. cat >> ==>

```
cat >> file1.txt
echo "Hello" >> file1.txt
echo "Linux" >> file1.txt
date >> file1.txt
cat notes.txt >> file1.txt
```

10. cp ==>

```
cp file1.txt backup.txt
cp file1.txt /tmp/
cp -r dir1 dir2
cp *.txt backup/
cp -v file1.txt file2.txt
```

11. mv (Move File)

```
mv file1.txt /tmp/
mv file1.txt Documents/
mv *.txt backup/
mv dir1 /tmp/
mv report.pdf Downloads/
```

12. mv (Rename File)

```
mv old.txt new.txt
mv test.c program.c
mv abc.doc xyz.doc
mv image.jpg photo.jpg
mv dir1 project
```

13. tac / rev

```
tac file.txt
tac log.txt
rev file.txt
rev words.txt
tac file.txt | rev
```

14. wc (Word Count)

```
wc file.txt
wc -l file.txt
wc -w file.txt
wc -c file.txt
wc -m file.txt
```

15. mkdir (Create Directory)

```
mkdir test
mkdir dir1 dir2 dir3
mkdir -p project/src
mkdir backup
mkdir docs
```

16. cd (Change Directory)

```
cd
cd ..
cd /home
cd ~/Documents
cd -
```

17. rm (Remove File/Directory)

```
rm file.txt
rm -i file.txt
rm -r testdir
rm -rf temp
rm *.txt
```

18. rmdir (Remove Empty Directory)

```
rmdir test
rmdir dir1
rmdir emptydir
rmdir folder1
rmdir sample
```

19. touch

```
touch file1.txt
touch file1 file2 file3
touch -a file.txt
touch -m file.txt
touch report.txt
```

20. sort

```
sort file.txt
sort -r file.txt
sort -n numbers.txt
sort -u file.txt
sort -k2 data.txt
```

21. uniq

```
uniq file.txt
uniq -c file.txt
uniq -d file.txt
uniq -u file.txt
sort file.txt | uniq
```

22. head

```
head file.txt
head -5 file.txt
head -20 log.txt
head -n 15 file.txt
head -c 50 file.txt
```

23. tail

```
tail file.txt
tail -5 file.txt
tail -20 log.txt
tail -f log.txt
tail -n 15 file.txt
```

24. factor

```
factor 10
factor 25
factor 100
factor 256
factor 999
```

25. alias

```
alias ll='ls -l'
alias la='ls -a'
alias cls='clear'
alias c='code'
alias update='sudo apt update'
```

26. unalias

```
unalias ll
unalias la
unalias cls
unalias c
unalias update
```

27. echo (Print Variables)

```
echo $HOME
echo $USER
echo $PATH
echo $SHELL
echo $PWD
```

28. ps (Processes)

```
ps
ps -e
ps -ef
ps aux
ps -u $USER
```

29. kill (Kill Process)

```
kill 1234
kill -9 1234
kill -15 1234
killall firefox
pkill chrome
```

30. env (Environment Variables)

```
env
env | less
env | grep HOME
env | grep PATH
env USER=ritik bash
```